Pore size influence on triboelectric charge density

"How does pore size influence triboelectric charge density in SiO2-MoS2 composites?" *Generated by LeapSpace*

Pore size appears to influence triboelectric charge density through a trade-off: larger or better-connected pores can raise charge generation by increasing contact area and trap sites, but overly large pores can reduce charge retention stability; for SiO_2–MoS_2 composites, the best-supported reading is that *moderate, well-distributed porosity* is more favorable than simply maximizing pore size. [46] [48] [71] [72] [73]

How pore size affects charge density

Pore-size change	Likely effect on triboelectric charge density	Evidence
Increase to a moderate level	**Raises charge density** by enlarging effective contact area, adding more frictional micro-regions, and creating more charge-trapping sites.	Larger pores and higher pore density are reported to improve triboelectric output and charge accumulation in porous TENGs [46] [48] [71] [73] [74] [75]
Too large / too open	Can **reduce charge stability** by making charge retention less secure and increasing the chance of loss.	The pore-size effect is described as beneficial up to an optimum, but excessively large pores may lower accumulation stability [71]
Smaller, more homogeneous pores	Can support **denser microscopic contact points** and more uniform charge accumulation, especially when the pore network is well controlled.	Smaller, more homogeneous pores are linked to localized charge accumulation and improved stability in porous triboelectric materials [71] [76]

What this means for SiO_2–MoS_2 composites

- In SiO_2–MoS_2 systems, the strongest support is for **engineered porosity plus a continuous shell/interface**, not for maximum pore size alone. A porous SiO_2 or mesoporous silica component can increase accessible surface area and trap density, while MoS_2 contributes strong charge trapping and interfacial charge regulation. [34] [77] [78] [79]
- The **MoS_2/SiO_2 interface** is itself important for charge trapping, so pore design matters most when it preserves that interfacial function rather than disrupting it. [34] [78] [80]
- The clearest design implication is that **pore size should be tuned to maximize contact electrification while preserving structural continuity and humidity resistance**. [34] [46] [71]

Bottom line

For **SiO_2–MoS_2 composites**, pore size most likely affects triboelectric charge density **nonlinearly**: moving from dense to moderately porous structures should increase charge density, but overly large pores can undermine charge retention. The most defensible target is **moderate, uniform, interconnected porosity** that boosts contact area and trapping without destroying the charge-blocking, humidity-resistant role of the SiO_2/MoS_2 architecture. [34] [46] [48] [71] [75]

Confidence level: **Medium** — the evidence strongly supports pore-size effects in porous TENG materials and in SiO_2/MoS_2-related charge-trapping systems, but it does not provide a direct pore-size sweep for one exact SiO_2–MoS_2 composite.

References

34. Kim, J.-K., Han, G.H., Kim, S.-W., (...), Baik, J.M. (2023) Electric-field-driven interfacial trapping of drifting triboelectric charges via contact electrification. *Energy and Environmental Science*. https://doi.org/10.1039/d2ee03114k

46. Mao, J., Wen, Y., Feng, X., (...), Seo, S. (2025) Multi-Sized porous silica/PDMS composite layer for enhanced performance in triboelectric nanogenerators. *Applied Surface Science*. https://doi.org/10.1016/j.apsusc.2025.162291

48. Chen, J., Guo, H., He, X., (...), Hu, C. (2016) Enhancing Performance of Triboelectric Nanogenerator by Filling High Dielectric Nanoparticles into Sponge PDMS Film. *ACS Applied Materials and Interfaces*. https://doi.org/10.1021/acsami.5b09907

71. Han, H., Cai, T., Ge-Zhang, S., (...), Tong, G. (2025) Wood-based triboelectric nanogenerators: From structural design to performance optimization and applications. *Materials and Design*. https://doi.org/10.1016/j.matdes.2025.115168

72. Gao, L., Li, J., Wang, Z., (...), Mu, X. (2022) A high performance triboelectric nanogenerator based on ordered doping technique for human-machine interaction sensing. *Nano Energy*. https://doi.org/10.1016/j.nanoen.2022.107025

73. Pharino, U., Sinsanong, Y., Pongampai, S., (...), Vittayakorn, W. (2021) Influence of pore morphologies on the mechanical and tribo-electrical performance of polydimethylsiloxane sponge fabricated via commercial seasoning templates. *Radiation Physics and Chemistry*. https://doi.org/10.1016/j.radphyschem.2021.109720

74. Han, J., Dai, X., Zhang, X., (...), Feng, W. (2026) Sensitivity and detection range-tailored self-powered sensing enabled by gradient porous nanocomposites for posture monitoring. *Nano Energy*. https://doi.org/10.1016/j.nanoen.2026.111819

75. Gulahmadov, O., Gahramanli, L., Muradov, M., (...), Trapalis, C. (2025) Porous polysiloxane MWCNT nanocomposites for high-performance and scalable triboelectric nanogenerators. *RSC Advances*. https://doi.org/10.1039/d5ra05894e

76. Butcha, S., Paiplod, P., Srisomwat, C., (...), Khemthong, P. (2025) Facile synthesis of SiO2/C composites derived from rice straw as high-performance anodes for lithium-ion batteries. *Diamond and Related Materials*. https://doi.org/10.1016/j.diamond.2025.112000

77. Han, B., Lv, X., Gao, X., Wang, J. (2018) Study on the effect of nano porous structure on the conductivity current of LDPE/SiO2 nano-composites. *Proceedings of the IEEE International Conference on Properties and Applications of Dielectric Materials*. https://doi.org/10.1109/ICPADM.2018.8401196

78. Wang, J., Xu, S., Li, G., (...), Hu, C. (2024) Regulation of deep and shallow hole/electron trap states and charge conducting behaviors of dielectric tribo-materials for maximizing retained charges. *Energy and Environmental Science*. https://doi.org/10.1039/d4ee03008g

79. Duan, Q., Peng, W., He, J., (...), Nie, S. (2023) Rational Design of Advanced Triboelectric Materials for Energy Harvesting and Emerging Applications. *Small Methods*. https://doi.org/10.1002/smtd.202201251

80. Zhou, Xinchen, Ruan, Hongcheng, He, Yuhui, (...), Zhai, Tianyou (2025) Ferroelectric transistor with tunable depolarization for artificial synapse with biphasic temporal response. *Applied Physics Letters*. https://doi.org/10.1063/5.0276048